Clock gating and an address-decoded memory for the cascade keyword spotter

Pedro Navarro (2132907)

I. What

The cascades B1 and B2 of R2 (Fig. 1) meet all seven group requirements, but their cost was estimated: the verifier time came from another SoC, and the PL draws 0.367 W whatever it computes, because the core busy-polls its mailbox and the 256 KiB BRAM reads on every cycle. R3 runs B1 and B2 on our fabric as one firmware and adds an idle mode in two steps. Step 1, clock gating, saves only 3.2 % (B1) and 3.6 % (B2). Step 2, an address-decoded memory, brings the PL power to 0.106 W and 0.105 W, −71 % and −72 %; every requirement still holds.

II. How

Step 1: clock gating (Fig. 2). Between requests the core stops on a store to a power register, and a BUFGCE stops its clock until the ARM's next request rings a doorbell. The logic is adapted from the streaming design's sleep branch; it needs no interrupts, and on an older bitstream the store simply returns.

Step 2: address-decoded memory (Fig. 3). Step 1 saves only 0.016 W, because the BRAM, 0.25 of the 0.28 W dynamic power, keeps reading. Read enables in the RTL cannot stop it: with byte writes, Vivado ties every enable of the bit-sliced layout to 1. In the address-decoded layout only the block that holds the address is on. Step 2 also reads the memory in 1 cycle instead of 8 (READ_WAIT 1): a verifier call takes 107 ms instead of 212 ms, which meets the 200 ms requirement.

Checks. Firmware and RTL were simulated (xsim) against the integer models: 117 requests with 13 verifier calls, bit-exact with and without gating and with identical cycle counts; with gating off, the RTL equals the previous design in every clock. Power is Vivado's estimate per state (CPU clock at its simulated duty, BRAM enables as built), combined with the simulated awake time; the board measurement is next.

III. Results

Fig. 5 shows where the power goes; Table II lists the cost of each step. Asleep, step 2 draws 0.102 W, mostly static power; the CPU sleeps 90.9 % (B1) and 94.2 % (B2) of the time. Fig. 4 and Table I compare the requirements with our earlier iterations: detection is unchanged, the measured compute time matches R2's estimate, and the power drops by about 70 %. Table III lists the predictions.

TABLE I
Group requirements: W0 from R1; B1 and B2 from R2 → R3 step 2 where R3 changes the value.

Requirement	Target	W0	B1	B2
Live "Sheila"	≥85 %	36.3 ×	87.8 ✓	85.8 ✓
False accepts per hour	≤2	2.86 ×	1.48 ✓	1.21 ✓
Other words accepted	≤0.2 %	0.03 ✓	0.15 ✓	0.15 ✓
Decision latency (ms)	≤500	134 ✓	171→170 ✓	178→178 ✓
Worst compute (ms)	≤200	0.35 ✓	108* →107 ✓	108* →107 ✓
Memory (KiB)	≤144	12.4 ✓	64.5 ✓	64.5 ✓
PL power (W)	≤1	0.367 ✓	0.367→0.106 ✓	0.367→0.105 ✓

*verifier time measured on another SoC (R2); R3 measures it on ours.

TABLE II
Cost and PL power of the steps (64 BRAM36 and 8 DSP in all; WNS = timing margin at 100 MHz).

Design	LUT	FF	BUFG	WNS (ns)	Idle (W)	B1 (W)	B2 (W)
Busy (as built)	3,243	2,831	1	+0.545	0.367	0.367	0.367
1: clock gating	3,302	2,915	2	+0.531	0.351	0.355	0.354
+ READ_WAIT 1	3,303	2,915	2	+0.479	0.350	0.352	0.351
2: + addr.-dec. BRAM	4,948	3,098	2	+0.366	0.102	0.106	0.105

IV. Pros and cons

+ PL power −71 % for B1 and B2 (0.106 and 0.105 W), all seven requirements met.
+ Bit-exact: detection is unchanged; all options are off by default, so earlier builds are reproduced.
+ Worst compute 107 ms on our fabric, well within 200 ms.
− +1,705 LUT and +267 FF; the timing margin (+0.37 ns) needs a timing-driven Vivado strategy.
− One BRAM block per port stays on while asleep; explicit banks could switch it off.
− Static power (0.080 W) now dominates; READ_WAIT 1 changes the cycle counts of earlier experiments.
− Vivado estimates only: the board measurement is still to do.

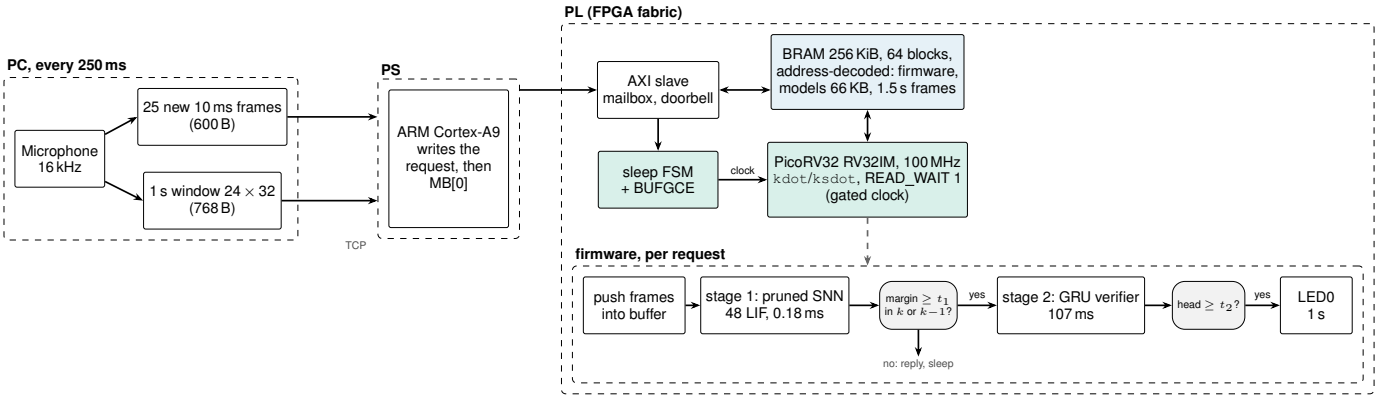


Fig. 1. The final design (B1 and B2 differ only in t_1). Every 250 ms the PC sends the newest 1 s window and the 25 new 10 ms frames. The core wakes, pushes the frames into a 1.5 s buffer and runs stage 1; if its margin reached t_1 in this request or the previous one, the GRU verifier scores the buffer, and a head score of at least t_2 (1 s hold-off) lights LED0. Then the core replies and sleeps (Fig. 2).

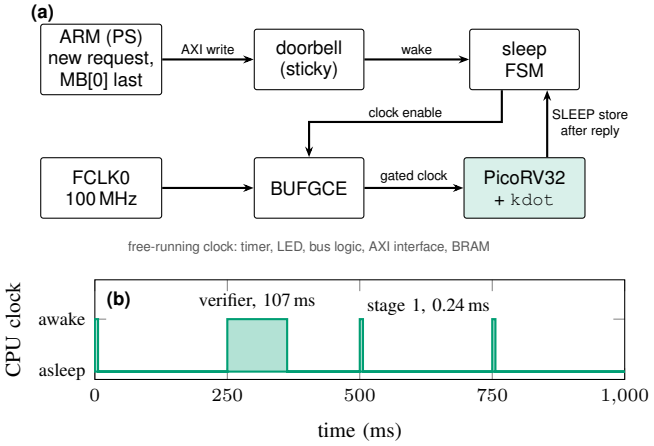


Fig. 2. Clock gating. (a) After each reply the core stores to a power register and stalls; the sleep FSM then stops its clock (one BUFGCE). The ARM writes the mailbox word MB[0] last; this rings a sticky doorbell that wakes the core, also if it rang just before the core went to sleep. (b) The CPU clock of step 2 over 1 s with one verifier call: awake 0.24 ms per request (drawn 6 ms wide) and 107 ms per verifier call; asleep 90.9 % (B1) and 94.2 % (B2) of the time.

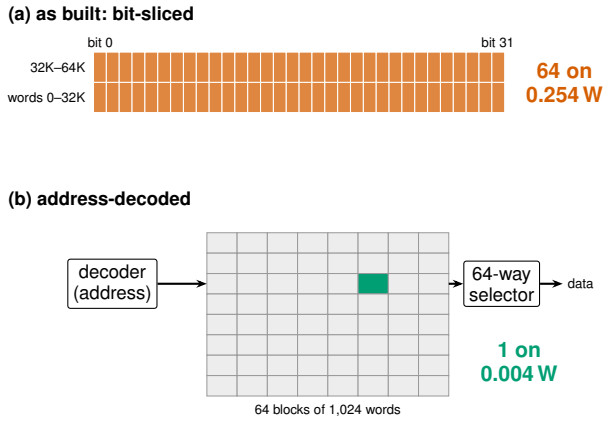


Fig. 3. The 256 KiB memory: 64 blocks of 36 Kbit, power while the CPU sleeps. (a) As built, each block holds one bit of every word, and with byte writes Vivado ties every enable to 1: all blocks read in every cycle. (b) Address-decoded, each block holds 1,024 whole words; only the block that holds the address is enabled, one per port.

TABLE III
PREDICTIONS WRITTEN BEFORE THE MEASUREMENTS, AND OUTCOMES.

	Prediction	Outcome
H1	verifier 19–25 M cycles at RW 8, worst >200 ms	✓21.2 M, 212 ms
H1′	10.3–11.2 M at RW 1, worst ≤200 ms	✓10.7 M, 107 ms
H2	bit-exact; same cycles with gating on/off	✓all sets
H3	idle 0.085–0.105 W	× step 1 (0.351); ✓step 2 (0.102)
H4	B1 ≤0.150, B2 ≤0.130 W	× step 1; ✓step 2
H5	gate alone ≤0.03 W; enables most of the rest	✓0.016 W; × enables
H6	asleep ≥90 / 93 % (B1 / B2)	✓90.9 / 94.2 %
H7	gating ≤+150 LUT, ≤+120 FF	✓+60, +84
H8	gating off = previous design	✓(+11 LUT)
H9	address-dec.: BRAM −80 %, ≤+150 FF	✓−98 %; × +183 FF
H10	READ_WAIT 1 saves only 1–5 mW	✓3.0 mW

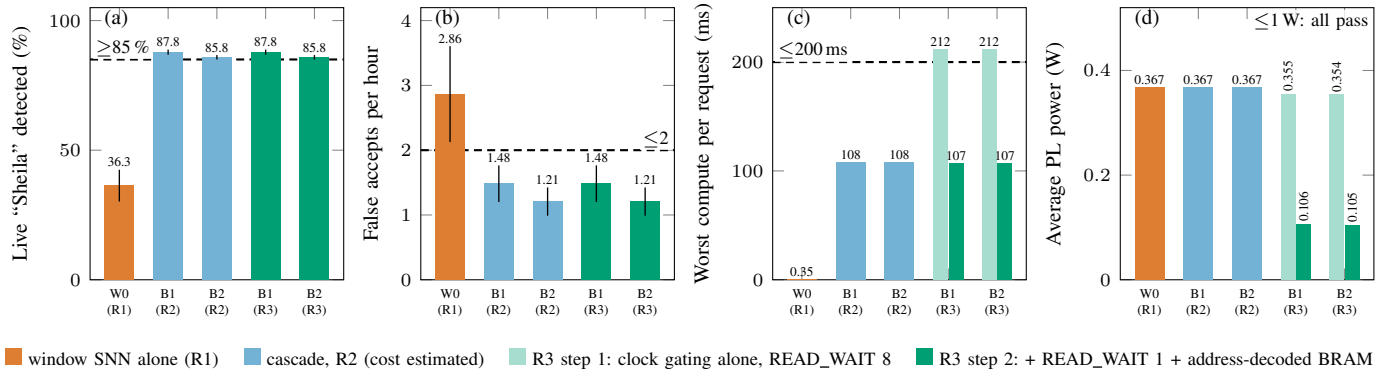


Fig. 4. Requirement metrics of our iterations (test set; mean ± sd over seeds). (a, b) Detection: R3 runs the R2 models bit-exactly, so these values carry over. (c) Worst compute per request: R2 assumed a verifier time measured on another SoC; on our fabric it is 212 ms with READ_WAIT 8 and 107 ms with READ_WAIT 1. (d) Average PL power with the verifier calls of R2 (B1 3,044, B2 1,908 per hour). The other words accepted, latency and memory requirements are in Table I.

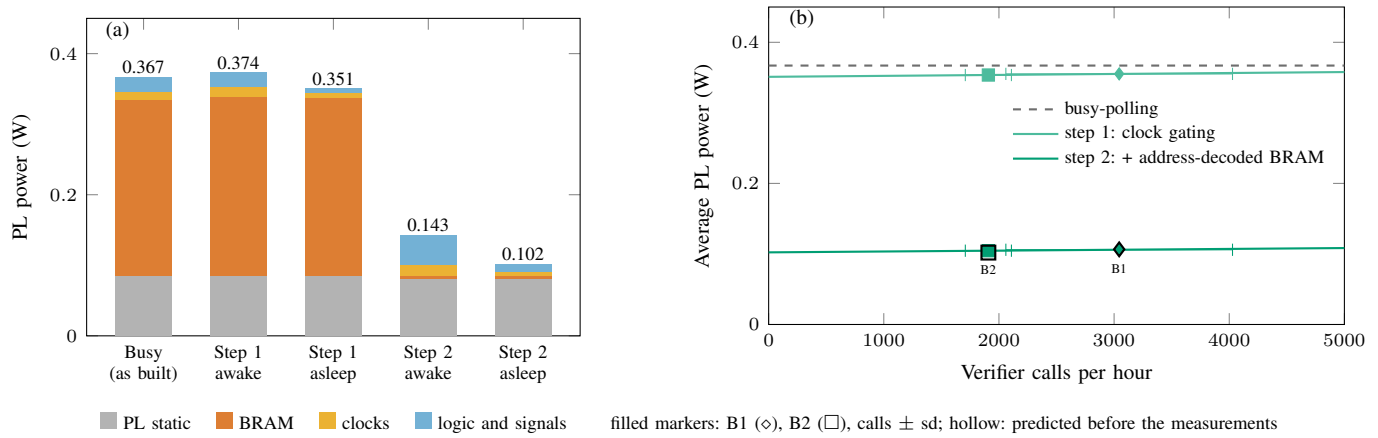


Fig. 5. PL power (Vivado). (a) Per state: step 1 gates the CPU clock (READ_WAIT 8); asleep it saves only 0.016 W, because the BRAM keeps reading. Step 2 (READ_WAIT 1, address-decoded BRAM) cuts the BRAM from 0.254 to 0.004 W; asleep, static power is most of what remains. (b) Average PL power against the verifier calls per hour (simulated awake time per request and per call): step 2 stays near its idle floor for any realistic call rate.